

Chain and Independent Food Service Business Performance According to Income and Neighborhood in Boston



Kamila Lim, Class of 2026
Data Science Major Capstone

Background and Aims

- Food health and safety are essential so that illness does not spread.
- Food chains often have standard procedures for sanitizing, food preparation, and handling.
- Investigating differences in food handling practices for chains across neighborhoods helps better understand whether there are trends between socioeconomic class, neighborhood, and food safety.
- Previous study in Philadelphia and British Columbia, Canada show that chain restaurants have fewer total violations per inspection than non-chain restaurants (Tung, 2018).
- Non-chain restaurants generally have a higher proportion of people living in poverty, and residents live in lower income with more Black and Hispanic residents near them (Leinwand et al., 2017).

Research Question

To what extent do chains perform differently (in terms of violations) than independent businesses in different neighborhoods based on income in Boston in 2024 to 2026?

Methods

- The original dataset consisted of one row per violation (*Food Establishment Inspections*, n.d.).
- National chains scraped from top grossing U.S. list (*Top 500 Chains*, n.d.). Brand chains are name chains not on the list. Institutional chains have the same owner but not the same name. Independents have one location (Figure 1).
- Data were spatially joined with census tracts (*TIGER/Line Shapefiles*, 2023).
- Income data joined and median income added to each row (U.S. Census Bureau, American Community Survey, 2024).
- Data were constrained to 2024-2026 and deduplicated so there is one row per inspection visit. Each visit has a column with the number of violations per visit.

Data Summary

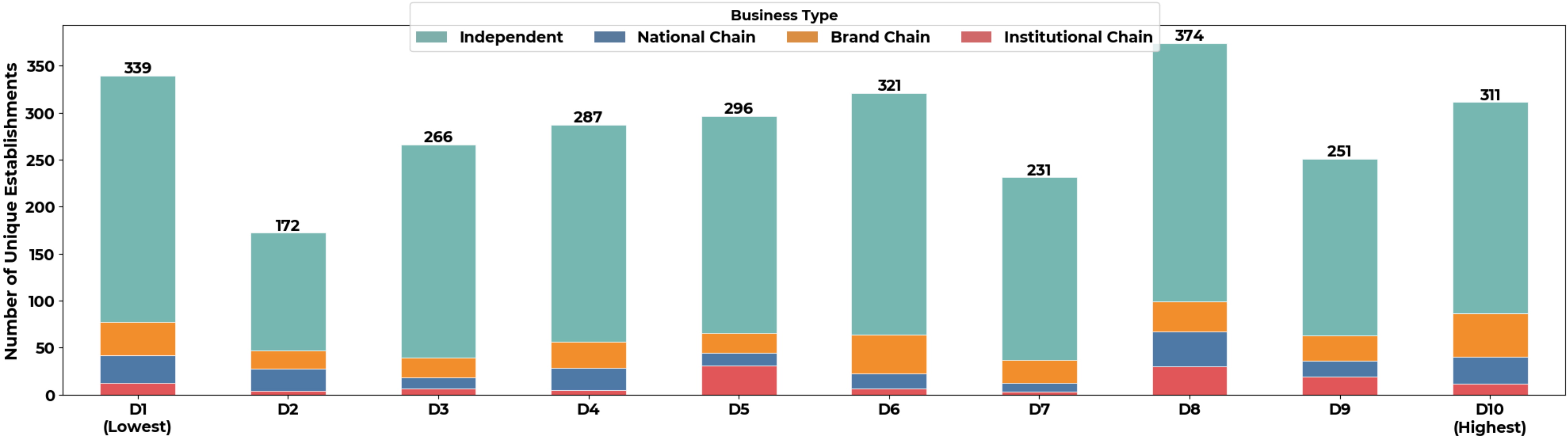


Figure 1: Unique Food Establishments by Business Type and Neighborhood Income Decile

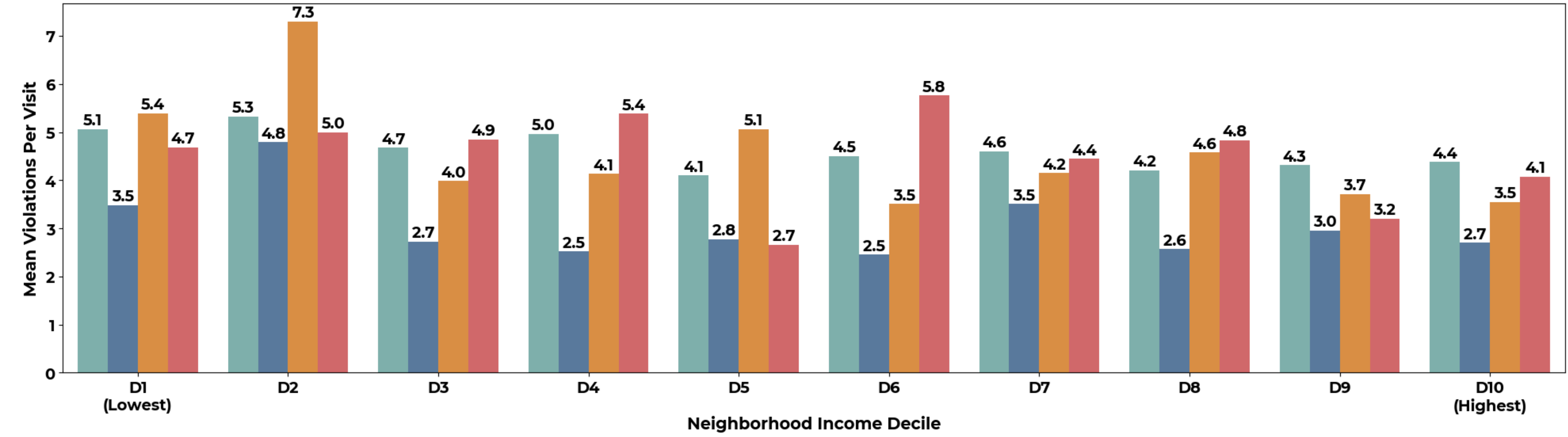


Figure 2: Mean Violations Per Visit by Business Type and Neighborhood Income Decile

Table 1: Negative Binomial Mixed Effects Interaction Model Results

Variable	Coefficient	Interpretations on log scale	Interpretations of violations	P-Value
Intercept	1.406	Intercept for independents	4.08 violations per visit	<0.001
National Chain	-0.392	Lower intercept than independents	32.5% fewer than independents	<0.001
Brand Chain	0.206	Higher intercept than independents	22.9% more than independents	0.009
Institutional Chain	-0.104	Intercept not different enough from independents	9.9% fewer than independents	0.386
Median Income	-0.007	Slope for independents	0.7% decrease with \$10k	0.072
National Chain x Median Income	0.002	Slope not different enough from independents	0.17% more with \$10k increase	0.771
Brand Chain x Median Income	-0.017	Smaller slope than independents	1.7% decrease with \$10k increase	0.007
Institutional Chain x Median Income	0.004	Slope not different enough from independents	0.4% increase with \$10k increase	0.660

Results and Discussion

- There are fewer violations in high income neighborhoods than in low-income areas for Brand Chains (Figure 2, Table 1).
- Negative binomial mixed effects model was used due to overdispersion in the Poisson.
- Random effects in mixed effects model include census tract and establishment. Fixed effects are analysis group and median income.
- Significant findings are as follows: National chains have 32.5% fewer violations than independents when income is \$0 (Table 1).
- Brand chains have 22.9% more violations than independents when income is \$0 (Table 1).
- For every \$10,000 increase in median household income, brand chain violations decrease by 1.7% (Table 1).
- Independents have 4.08 violations compared to average of 4.45 violations (Table 1).
- Fixed effects explain 2.8% of the variance (Marginal R²) and the full model explains 39.9% of the variance (conditional R²).

Limitations

- There is potentially temporal and spatial correlation. Two tracts next to each other may have more similarities than one far away. Time isn't accounted for in the model.
- Independents comprise most inspection visits. As such, it is more difficult to make accurate predictions for chains (Figure 1).
- A few chains dominate the National Chain category, meaning that these few chains' outcomes represent the entire group.
- Potential clustering based on business isn't considered – a potential independence issue. One Dunkin Donuts may be more like another.

References

- Food Establishment Inspections. (n.d.). [Tabular]. Inspectional Services Department. Retrieved <https://data.boston.gov/dataset/food-establishment-inspections>
- Leinwand, S. E., Glanz, K., Keenan, B. T., & Branas, C. C. (2017). Inspection Frequency, Sociodemographic Factors, and Food Safety Violations in Chain and Nonchain Restaurants, Philadelphia, Pennsylvania, 2013-2014. *Public Health Reports*, 132(2), 180-187. <https://doi.org/10.1177/0033354916687741>
- TIGER/LineShapefiles (2023) [Dataset]. <https://www.census.gov/cgi-bin/geo/shapefiles/index.php>
- Top 500 Chains. (n.d.). Restaurant Business. Retrieved <https://www.restaurantbusinessonline.com/top-500-2020>
- Tung, J., BCIT School of Health Sciences, Environmental Health, & Heacock, H. (2018). Critical food safety violations in Surrey: relationship to community median household income and restaurant type. *BCIT Environmental Public Health Journal*. <https://doi.org/10.47339/ephj.2018.57>
- U.S. Census Bureau, American Community Survey. (2024). S1901: Income in the Past 12 Months (in 2024 Inflation-Adjusted Dollars) [Dataset]. <https://data.census.gov/all?q=S1901>